Production-Grade 5G Core

Phase-wise Build Strategy – AMF, SMF, UPF, PCF, UDM first

Version 0.1 (draft for review) • 2026-09-30

Executive summary

This document lays out how we build our own commercial-grade 5G Core (5GC) from scratch. The end state is a production core with all 11 network functions (NFs). We start with five – **AMF, SMF, UPF, PCF, UDM** – because they form the minimum set that carries a real UE from registration to user-plane traffic.

Four principles drive every decision below:

- **Clean-room, licence-safe.** Implement from the free 3GPP specifications. Open5GS (AGPL-3.0) and other open-source cores are used only as *black-box behavioural references and interop peers* – never as a source of copied code.
- **Spec-first, test-first.** Every procedure maps to a 3GPP clause and a test. Progress is measured in procedures passing against independent implementations, not lines of code.
- **Independent validation.** Our ORAN emulator (UE/gNB) is validated against Open5GS. Now the flow reverses: open-source UE + gNB (UERANSIM, srsRAN) validate *our* core. Open5GS stays as a cross-check oracle.
- **Production from day one.** Observability, config, TLS, packaging and CI are built in Phase 0, not bolted on at the end.

Headline plan

Phase	Focus	Exit gate
0	Foundations: repo, legal, specs, shared libs	Libs pass spec test vectors; CI green
1	UDM (+ lite UDR/AUSF shims)	5G-AKA vectors and UE-auth API verified
2	PCF	SM/AM policy APIs interop with mock SMF/AMF
3	UPF	PFCP + GTP-U forwarding at target throughput, vs. a PFCP test client
4	SMF	PDU session lifecycle against our UPF/PCF/UDM
5	AMF	UERANSIM registers a UE and pings via our core
6	Integration & validation of the 5-NF core	Conformance matrix ≥95% pass; soak 72 h
7	Remaining 6 NFs (NRF, AUSF, UDR, NSSF, NEF, +1)	Shims removed; full SBA discovery
8	Production hardening & HA	Perf/HA/security targets met
9	Productisation & release	GA-candidate, licence audit clean

1. Scope, assumptions and open decisions

1.1 What the five NFs do

NF	Role	Key interfaces	Main 3GPP specs
AMF	Registration, connection & mobility, NAS security, N2 termination toward gNB	N1/N2 (NAS, NGAP), N8 (UDM), N11 (SMF), N15 (PCF), N12 (AUSF)	TS 23.501/502, 24.501, 38.413, 29.518
SMF	PDU session management, UE IP allocation, UPF selection/control, QoS flow set-up	N4 (PFCP), N11, N7 (PCF), N10 (UDM)	TS 23.501/502, 24.501, 29.502, 29.244
UPF	Packet forwarding, GTP-U tunnel termination, QoS enforcement, usage reporting	N3 (gNB), N4 (SMF), N6 (DN), N9	TS 29.244, 29.281, 23.501
PCF	Policy and charging rules: session, access & mobility, UE policy	N7, N15, N36 (UDR)	TS 23.503, 29.512, 29.507, 29.525
UDM	Subscriber data, authentication credential handling, SUCI de-concealment, registration state	N8, N10, N13 (AUSF), N35 (UDR)	TS 23.501, 29.503, 33.501

1.2 Hidden dependency: five NFs are not enough to run a network

A real UE registration needs **NRF** (discovery), **AUSF** (5G-AKA) and **UDR** (data store) in addition to the five. To stay focused we will build deliberately small **shims** (`shims/`) that expose only the required service operations, backed by static config or a simple DB. The shims are scaffolding: each is deleted in Phase 7 when its real NF lands. Shims must never contain logic we intend to ship.

1.3 Assumptions (please confirm or correct)

- Target baseline: **3GPP Release 16** features (SA mode, non-roaming), with Release 17 items added selectively. Non-standalone/4G interworking is out of scope.
- The “11 NFs” are taken as: AMF, SMF, UPF, PCF, UDM (first five) plus NRF, AUSF, UDR, NSSF, NEF and one of SCP/BSF. **Please confirm the 11th.**
- Deployment target: containers on Kubernetes (plus bare-metal UPF for performance), PLMN test codes 001/01 during development.

1.4 Decisions needed before Phase 0 ends

#	Decision	Recommendation	Why
D1	Implementation language	**C** (C17, GCC/Clang) for all NFs, with mandatory memory-safety controls (ADR-0013)	Widest toolchain and hiring base, direct access to AF_XDP/DPDK for the UPF, no GC pauses. The cost is memory safety in network-facing parsers, so the build enforces -Werror and hardening flags, ASan/UBSan on every test run, fuzzing of every decoder, a checked buffer type and static analysis. Licensing is language-independent.
D2	Product licence	Proprietary / commercial (our choice), enforced by a permissive-only dependency policy	Keeps the product closed-source capable; see Section 2.
D3	Release baseline	Rel-16 (TS versions pinned in docs/specs)	Stable, widely supported by UERANSIM/srsRAN.
D4	Subscriber store	PostgreSQL behind a UDR-style API (SQLite in dev)	Avoids MongoDB’s SSPL-licence question.
D5	Data-plane technology	Userspace first, then AF_XDP/eBPF; DPDK optional	Gets correctness first, performance second.

2. Licensing and IP strategy (launch-as-a-product safe)

Licence risk is the biggest non-technical risk of a telecom product. The rules below are mandatory from the first commit. **This is an engineering policy, not legal advice – have IP counsel review it before launch.**

2.1 How we may use each reference

Project	Licence	Use in our project	Not allowed
Open5GS	AGPL-3.0	Run unmodified binaries as a separate process for interop and behaviour comparison. Read its public behaviour/logs.	Copying or translating source, config schemas or generated code (e.g. its ASN.1 output) into ours; linking it; shipping it in our product.
free5GC	Apache-2.0	Architectural reference; peer for interop. Code may technically be reused with attribution, but we choose not to for clean provenance.	Mixing code without NOTICE/attribution.
UERANSIM	AGPL-3.0 (dual-licensed with a commercial licence)	Test tool only (UE + gNB), run in CI as an external process.	Linking, embedding or redistributing in our product.
srsRAN Project / srsUE	AGPL-3.0	Second RAN test peer (with SDR or ZMQ virtual radio).	Same as above.
OpenAirInterface 5G core	OAI Public Licence v1.1 (patent clause, non-commercial bias)	Avoid; not needed.	Any use in or near the product.
3GPP specs / OpenAPI YAML	Free to download; 3GPP copyright terms	Primary source. We implement from these. Generated API stubs are OK under 3GPP terms – confirm with counsel.	Redistributing the spec documents themselves.

2.2 Engineering controls

- **Clean-room rule:** engineers implement from the specs; nobody pastes code from AGPL/GPL projects. Reading them for behaviour clarification is allowed but any design note must cite the 3GPP clause, not the repo.
- **Provenance log:** each module lists its governing spec clauses in its header and in docs/specs. PRs touching protocol logic must cite the clause.
- **Dependency policy:** only MIT / Apache-2.0 / BSD / ISC / MPL-2.0 (file-level) dependencies. No GPL/AGPL/SSPL/BUSL. Enforced in CI (scripts/c_lint.sh against docs/legal/THIRD_PARTY.md) with an SBOM (CycloneDX) generated per build.
- **Own ASN.1 pipeline:** NGAP (TS 38.413) codec generated from the 3GPP ASN.1 with a permissively-licensed compiler or our own generator; never reuse Open5GS's generated C.
- **Own crypto code path:** use an audited permissive library (OpenSSL 3, Apache-2.0) for AES, SHA, ECIES; implement Milenage / 5G key hierarchy ourselves and verify against spec vectors (TS 35.208, TS 33.501 Annex).
- **Patents (SEPs):** 5G has standard-essential patents held by third parties, independent of copyright. Open-source origin does not remove that exposure. Budget a patent-landscape / FRAND review before commercial launch.
- **Test tools are quarantined:** UERANSIM/srsRAN/Open5GS live only in test containers under `tests/` and `deploy/compose`, pulled at test time, never in release artifacts.

3. Target architecture and engineering foundations

3.1 Repository layout

```
5G_core/
  README.md                 short intro, pointer to docs/
  docs/
    README.md               documentation index, conventions, how to build/run
    update.md               append-only progress / build log  (see 3.3)
    strategy/               this PDF
    adr/                    architecture decision records (NNNN-title.md)
    specs/                  spec index: clause -> module -> test mapping
  libs/                     shared libraries (no NF-specific logic)
    sbi/                    HTTP/2 + JSON SBI server/client, OAuth2, OpenAPI models
    nas/  ngap/  pfcp/  gtpu/     protocol codecs
    crypto/                 Milenage, 5G-AKA, KDFs, SUCI ECIES, NEA/NIA
    config/  telemetry/     config loading; logs, metrics, traces
    testkit/                mocks, packet builders, fixtures
  nf/
    amf/  smf/  upf/  pcf/  udm/         the five target NFs
  shims/                    temporary: nrf-lite, ausf-lite, udr-lite (deleted in Phase 7)
  tests/
    unit/  integration/  e2e/  conformance/  fixtures/
  deploy/                   docker/  compose/ (UERANSIM, Open5GS oracle)  k8s/
  configs/                  sample YAML per NF
  tools/  scripts/          dev tooling, pcap helpers, licence/SBOM checks
```

3.2 Architectural rules

- **One binary per NF**, stateless where possible; state in an external store so NFs can scale and restart (needed for Phase 8 HA).
- **Shared code only via libs/**; NFs never import each other. Inter-NF contracts are the 3GPP OpenAPI schemas (SBI) and PFCP/NGAP/NAS (non-SBI).
- **Uniform NF skeleton:** config → telemetry → SBI server → NRF registration (static in shim mode) → health/readiness endpoints.
- **Observability built in:** structured logs with UE/session correlation IDs, Prometheus metrics, OpenTelemetry traces, pcap capture hooks.
- **Security built in:** TLS 1.3 / mTLS on SBI, OAuth2 access tokens (TS 33.501 §13) behind a flag in early phases, secrets never in config files.
- **CI from Phase 0:** build, unit tests, licence scan, SBOM, fuzzing of every decoder (NAS, NGAP, PFCP, GTP-U, SBI JSON), container build.

3.3 Documentation system

`docs/README.md` is the stable entry point: architecture summary, build/run instructions, conventions and an index of all docs. `docs/update.md` is an **append-only, newest-first log**. Every working session or merged milestone adds one entry with: date, phase, what was built, spec clauses covered, tests added and their status, known gaps, and next steps. A running status table at the top of update.md shows per-NF progress. ADRs capture every decision in 3.1 with context and consequences.

4. Validation strategy

4.1 Test pyramid

Layer	What	Tools
Unit	Codecs round-trip; crypto against published vectors; state machines	Native test framework, property-based tests
Fuzz	Every network-facing decoder, continuously	libFuzzer (clang) plus a gcc replay driver in ctest, AFL++
Component	One NF against scripted mocks of its peers (e.g. SMF vs mock UPF + mock PCF)	libs/testkit mocks; PFCP/SBI replay
Interop	Our NF swapped into an Open5GS/free5GC network, or vice versa, one NF at a time	Open5GS as oracle in compose
End-to-end	Real UE + gNB simulators drive registration, PDU session, traffic, deregistration	UERANSIM (primary), srsRAN + ZMQ (secondary)
Conformance	Per-procedure pass/fail matrix against spec traces; pcap diffing vs Open5GS for the same scenario	Wireshark/tshark dissectors, scripts/
Non-functional	Throughput, latency, soak, failure injection, security scans	iperf3, TRex, chaos tooling, nmap/ZAP

4.2 Using your ORAN emulator

The ORAN emulator's UE/gNB currently proves its correctness against Open5GS. Once our core runs, it becomes a **third, independent peer**. If UERANSIM, srsRAN and the emulator all agree with our core, and the emulator also agrees with Open5GS, then a mismatch is localised quickly: same scenario, three RAN clients, two cores. Run the same scenario scripts against both cores and diff the pcaps.

4.3 E2E scenario catalogue (grows with phases)

ID	Scenario	Needs
S1	Initial registration with SUCI, 5G-AKA, NAS security mode	AMF, UDM (+AUSF-lite)
S2	PDU session establishment (IPv4), ping through N6	+ SMF, UPF, PCF
S3	Deregistration (UE and network initiated), PDU session release	S2
S4	Service request after idle (RRC release / paging)	AMF, SMF, UPF
S5	Multiple UEs and multiple PDU sessions, QoS flows (5QI 9 + GBR)	SMF, PCF, UPF
S6	Xn / N2 handover, multi-gNB	AMF, SMF, UPF
S7	Failure cases: bad MAC, unknown subscriber, reject causes, UPF restart	All
S8	IPv6 / IPv4v6, multiple DNNs, slices (S-NSSAI)	SMF, UPF (NSSF in Phase 7)

Phase 0 – Foundations

Goal. Everything every NF needs, plus the legal and process groundwork, so the five NFs are built on one consistent base.

Scope / deliverables

- Repo scaffold, toolchain pin, CI pipeline, lint/format, container builds, licence scan + SBOM.
- ADRs D1–D5 decided and recorded; spec baseline (Rel-16 TS list and versions) in docs/specs; clause-to-module traceability table.
- **libs/sbi:** HTTP/2 server/client, JSON (de)serialisation, OpenAPI-driven model generation from 3GPP YAML, ProblemDetails, retry/timeouts, TLS, OAuth2 token hooks.
- **libs/nas** (TS 24.501 5GMM/5GSM), **libs/ngap** (TS 38.413, own ASN.1 pipeline), **libs/pfcp** (TS 29.244), **libs/gtpu** (TS 29.281).
- **libs/crypto:** Milenage, 5G-AKA / 5G key hierarchy (KAUSF→KSEAF→KAMF→KgNB), SUCI ECIES profiles A and B, NEA0-3 / NIA0-3.
- **libs/config, telemetry, testkit:** YAML config with validation, logging/metrics/tracing, packet builders, mock-peer harness.
- Compose environment bringing up UERANSIM and Open5GS (as oracle) so baseline traces exist from day one; capture golden pcaps of S1–S3 from Open5GS.

Spec procedures covered

- Not applicable (infrastructure).

Validation

- Codec round-trip tests on captured Open5GS/UERANSIM pcaps; decoders fuzzed for a fixed time budget.
- Crypto matches published vectors in TS 35.208 and TS 33.501 annexes.

Exit criteria

- CI green on all libs; licence scan clean; fuzzing running.
- Golden pcaps stored in tests/fixtures.
- ADRs signed off.

Key risks

- ASN.1 toolchain choice can be a time sink – time-box and decide early.
- Scope creep into “perfect” libs: implement Rel-16 SA subset needed for S1–S8 only.

Phase 1 – UDM (+ UDR-lite, AUSF-lite)

Goal. A working subscriber-data and authentication backbone, exercised by mock consumers, before any RAN exists.

Scope / deliverables

- **UDM services:** Nudm_UEAuthentication (generate auth data, auth-event confirm), Nudm_SubscriberDataManagement (AM, SM, NSSAI data; subscribe/notify), Nudm_UEContextManagement (AMF/SMF registration).
- SUCI de-concealment (home network private key handling, profiles A/B) and SUPI resolution.
- Subscriber provisioning: REST + CLI for IMSI, K/OPc, AMF field, SQN, DNN/slice profile. Secure storage of K/OPc (envelope encryption hook for later HSM/KMS).
- SQN management and resynchronisation (AUTS handling).
- **shims/udr-lite** (Nudr_DataRepository subset) and **shims/ausf-lite** (Nausf_UEAuthentication 5G-AKA subset) strictly as needed.

Spec procedures covered

- TS 33.501 §6.1 (5G-AKA), §6.12 (SUCI); TS 29.503; TS 29.504/29.505 (UDR subset); TS 29.509 (AUSF subset).

Validation

- Auth vectors compared with Open5GS UDM output for identical K/OPc/RAND/SQN.
- Mock AMF/AUSF clients drive all UDM operations; negative tests (unknown SUPI, bad SUCI scheme, SQN mismatch).

Exit criteria

- Bit-exact vectors match the oracle and spec.
- Provisioning round-trip works; 1,000 auth req/s on one core-class instance (baseline).

Key risks

- Key handling is security-critical: code review + threat note before exit.
- Keep shims minimal or they become legacy we must migrate.

Phase 2 – PCF

Goal. Policy decisions available to SMF and AMF with subscriber-linked policy data.

Scope / deliverables

- **Npcf_SMPolicyControl** (create/update/delete; PCC rules, QoS (5QI, ARP, AMBR), session rules) and **Npcf_AMPolicyControl**.
- Policy store via UDR-lite; default policy profile plus per-subscriber / per-DNN overrides; policy templates in config.
- Policy-update notifications toward SMF (Npcf_SMPolicyControl_UpdateNotify).
- Hooks reserved (not built) for UE policy (UCU), AF-influenced traffic (N5) and usage monitoring – Phase 7/8.

Spec procedures covered

- TS 23.503; TS 29.512 (SM policy); TS 29.507 (AM policy); TS 29.519 (policy data in UDR).

Validation

- Component tests with mock SMF/AMF; policy decision tables tested against spec examples.
- Interop: Open5GS SMF ↔ our PCF, with identical QoS outcome in the SMF trace.

Exit criteria

- Open5GS SMF accepts our PCF decisions for S2-style sessions.
- Policy change mid-session reaches the SMF.

Key risks

- Behavioural differences in optional IEs between implementations – capture in update.md as they surface.

Phase 3 – UPF

Goal. A correct, then fast, user plane that any spec-compliant SMF can drive.

Scope / deliverables

- **PFCP (N4):** association setup/heartbeat/release, session establishment/modification/deletion/report; PDR, FAR, QER, URR, BAR; UE IP pool hooks; SDF filter matching.
- **GTP-U (N3/N9):** encapsulation/decapsulation, echo, error indication, end-marker, extension headers (PDU Session Container, QFI).
- **N6:** TUN/TAP and routed interface to the data network; NAT/masquerade as config option.
- QoS enforcement: session-AMBR, MBR/GBR per QER, QFI marking. Usage reports (volume/time) for future charging.
- Stage A: userspace datapath for correctness. Stage B: AF_XDP/eBPF fast path; multi-queue, RSS, CPU pinning. Benchmarks published in docs.

Spec procedures covered

- TS 29.244; TS 29.281; TS 23.501 §5.8, §5.7 (QoS).

Validation

- Driven by a scripted PFCP client and (interop) Open5GS SMF ↔ our UPF with UERANSIM traffic.
- Packet-level tests: golden GTP-U frames in/out; fuzzing of PFCP and GTP-U parsers.
- iperf3/TRex throughput and latency at increasing UE/session counts.

Exit criteria

- Open5GS SMF + UERANSIM: ping and iperf through our UPF.
- Stage A: ≥ 1 Gbps single host; target for Stage B set as an ADR (e.g. 10 Gbps).

Key risks

- Performance work can swallow effort – gate it behind correctness.
- Windows is a dev host only; UPF development and performance testing need Linux (WSL2/VM).

Phase 4 – SMF

Goal. Complete PDU session lifecycle control across UDM, PCF, UPF and (mock) AMF.

Scope / deliverables

- **Nsmf_PDUSession:** Create/Update/Release SM Context (from AMF), N1N2 message transfer, status notify.
- 5GSM NAS handling via AMF (establishment accept/reject, modification, release) using libs/nas.
- **N4 control:** UPF association management, UPF selection (DNN, slice, load/static), rule generation from PCC rules (PDR/FAR/QER).
- IP address management (IPv4, IPv6 prefix, IPv4v6), DNN/DNS/MTU config, session AMBR, QoS flow mapping (default + dedicated).
- UDM subscription retrieval (SM data) and PCF SM policy integration; session/SM-context persistence to external store.
- Handling of UPF failure/restart, N4 session recovery; idempotent retries.

Spec procedures covered

- TS 23.502 §4.3 (PDU session procedures); TS 29.502; TS 24.501 (5GSM); TS 29.244.

Validation

- Mock AMF drives full lifecycle; real UPF (ours) + real PCF/UDM (ours).
- Interop: Open5GS AMF ↔ our SMF ↔ Open5GS UPF and our UPF (all combinations) with UERANSIM.

Exit criteria

- S2, S3, S5 (partial) pass with Open5GS AMF in front; pcap diff on N4 shows no semantic deviations.
- Crash/restart test: sessions recovered or cleanly released.

Key risks

- QoS/PCC-to-PFCP mapping is the most intricate logic – build a rule-mapping test table from spec early.

Phase 5 – AMF

Goal. The last and most complex of the five: real UE and gNB attach to our core end to end.

Scope / deliverables

- **N2/NGAP:** SCTP endpoint, NG Setup, gNB/UE context, Initial UE Message, UL/DL NAS transport, Initial Context Setup, UE Context Release, PDU Session Resource Setup/Modify/Release, Paging, Reset, Error Indication.
- **NAS (5GMM):** Registration (initial/mobility/periodic), Authentication, Security Mode (integrity + ciphering), Identity, Configuration Update, Service Request, Deregistration, Notification; GUTI allocation; 5G-TMSI paging.
- UE context and registration-area management, tracking-area lists, allowed NSSAI from config/UDM, 5GMM timers.
- SBI: Namf_Communication, Namf_EventExposure (basic), calls to AUSF (shim), UDM, PCF (AM policy), SMF.
- Handover: Xn-based path switch, then N2-based handover (S6). Multi-gNB support.
- Stateless-ready design: UE context externalised for later HA; SCTP multi-homing config.

Spec procedures covered

- TS 23.502 §4.2 (registration, service request, deregistration), §4.9 (handover); TS 24.501; TS 38.413; TS 29.518; TS 33.501 §6 (security procedures).

Validation

- Replay of captured Open5GS/UERANSIM NAS/NGAP traces as decoder and state-machine tests.
- UERANSIM gNB + UE: S1, S2, S3, S4; then srsRAN and the ORAN emulator.

Exit criteria

- Phase-5 milestone: **UERANSIM UE registers, establishes a PDU session and pings 8.8.8.8 through our 5 NFs (plus shims).**
- S1–S4 green on UERANSIM and on the ORAN emulator.

Key risks

- NAS/NGAP edge cases and timer behaviour are where interop bugs hide – use traces and fuzzing heavily.
- Security-mode/ciphering errors look like generic failures – invest in good diagnostics early.

Phase 6 – Integration & validation of the 5-NF core

Goal. Prove the five-NF core works as a system, independently of any single-phase test, and establish the regression suite we keep forever.

Scope / deliverables

- Full S1–S8 on UERANSIM, srsRAN (ZMQ) and the ORAN emulator; multi-UE, multi-session scale tests (100 → 1,000 → 10,000 simulated UEs).
- Conformance matrix: each procedure → clause → test → status, published in docs/specs; differential pcap analysis against Open5GS.
- 72-hour soak, memory-leak and FD-leak checks, failure injection (kill NF, drop N4/N11, SCTP flaps).
- Bug-burn-down and performance baseline report; decision checkpoint before Phases 7–9.

Spec procedures covered

- Whole-system view of TS 23.502 procedures in scope.

Validation

- As above.

Exit criteria

- ≥ 95% of in-scope scenarios pass on all three RAN peers; no P1 bugs open; soak clean.
- Baseline report stored in docs/.

Key risks

- Differences between RAN simulators' behaviours – treat the spec as the tiebreaker, record rulings in ADRs.

Phase 7 – Remaining NFs and shim removal

Goal. Complete the 11-NF core and retire all temporary code.

Scope / deliverables

- **NRF** (Nnrf_NFManagement, NFDiscovery, AccessToken; heartbeat/status notify) – first, since every NF then switches from static config to discovery.
- **AUSF** and **UDR** replace the shims (full 5G-AKA, EAP-AKA' optional; full Nudr data sets, PostgreSQL-backed).
- **NSSF** (slice selection, NSSAI availability), **NEF** (northbound exposure, AF traffic influence, monitoring events, CAPIF-ready hooks).
- **11th NF (SCP or BSF, to confirm)**: SCP for indirect communication / routing, or BSF for PCF binding.
- PCF extensions: UE policy, N5/AF interaction, usage monitoring. Optional: UDM/AMF event exposure completion.

Spec procedures covered

- TS 29.510, 29.531, 29.509, 29.504–29.505, 29.522, 29.521 and related.

Validation

- Regression: the full Phase-6 suite must still pass, unchanged, after each shim is replaced.

Exit criteria

- All 11 NFs deployed; `shims/` deleted; multi-slice scenarios (S8) green.

Key risks

- Regressions while swapping shims – keep the same external APIs and run the Phase-6 suite on every PR.

Phase 8 – Production hardening

Goal. Turn a working core into an operable, secure, scalable product.

Scope / deliverables

- **HA and scale:** stateless NF replicas, external session/UE stores (PostgreSQL / Redis-class), SCTP and N4 failover, rolling upgrades, horizontal autoscaling.
- **Security:** mTLS between all NFs, OAuth2 with NRF tokens enforced, secrets management (KMS/HSM for K/OPc and home-network keys), threat model, pen-test, dependency and container scanning, SBOM and signed artifacts.
- **Operations:** Helm charts/operators, config hot-reload, health probes, Prometheus + Grafana dashboards, alert rules, structured audit logs, runbooks.
- **Performance:** UPF tuning (10–100 Gbps targets by ADR), control-plane registrations/sec and sessions/sec targets, latency SLOs.
- **Operator features:** charging hooks/CDR export (Nchf integration point), lawful-intercept interface design (jurisdiction-dependent), subscriber provisioning API and UI/CLI, multi-PLMN.
- **Commercial testers:** run the conformance scenarios with a commercial core-test tool (Keysight/Spirent class) or a partner lab.

Spec procedures covered

- TS 33.501 (security architecture), TS 28.xxx (management) as applicable.

Validation

- Chaos/failure suite, 7-day soak, load tests to target, external pen-test with remediation.

Exit criteria

- SLOs and capacity targets met and documented; no critical/high security findings open; upgrade and rollback tested.

Key risks

- Lawful intercept and charging requirements depend on target markets – scope them with stakeholders early.

Phase 9 – Productisation and release

Goal. Package, document, license-audit and ship a release candidate.

Scope / deliverables

- Versioning and release process, packaged containers/Helm, installation and upgrade guides, operator/admin documentation, API reference.
- Final third-party licence audit (automated + manual), NOTICE file, SBOM, patent/FRAND review outcome recorded.
- Support tooling: diagnostics bundle, pcap capture, log redaction.
- Pilot / field trial plan with a lab or partner network; feedback loop into the backlog.

Spec procedures covered

- n/a

Validation

- Release-candidate regression on full matrix.

Exit criteria

- GA-candidate tagged; licence audit signed off by counsel; pilot plan agreed.

Key risks

- Late discovery of licence or patent issues – mitigated by running checks from Phase 0.

5. Program-level risks and mitigations

Risk	Impact	Mitigation
Licence contamination (accidental copy from AGPL/GPL code)	Could force open-sourcing or block launch	Clean-room policy, PR clause citations, CI licence scan, periodic code-similarity scan before release
SEP / patent claims	Royalty or injunction exposure	Patent-landscape review with counsel before Phase 9; budget for FRAND licences
Interop gaps between simulators	False failures or false passes	Three RAN peers + Open5GS oracle; spec is the tiebreaker; ADR for each ruling
Protocol complexity underestimated (AMF/NAS)	Delivery slip	Trace replay tests early, fuzzing, time-boxed spikes, re-plan at the Phase-6 checkpoint
SNOW 3G / ZUC algorithm licensing (NIA1/NEA1 are mandatory for an AMF)	Cannot implement from 3GPP documents; algorithm specs are ETSI SAGE documents under licensing conditions	Decide early: obtain the SAGE specifications and licence terms, budget for them, and review with counsel; AES-based NIA2/NEA2 are unaffected
Shim creep	Legacy code in product	Shims isolated in shims/, API-identical to real NFs, deletion is a Phase-7 exit criterion
UPF performance shortfall	Product not competitive	Correctness first; performance budget per ADR; Linux perf lab from Phase 3
Key-person and spec knowledge concentration	Bus factor	docs/specs mapping, ADRs, update.md log, pair reviews
Windows dev host	Build/test friction (SCTP, TUN, XDP)	Use WSL2/Linux VM or a Linux dev box for UPF/AMF integration work

6. Definition of done (per NF and per phase)

- All in-scope procedures implemented with spec clause references; unit + component + interop tests passing in CI.
- Fuzzing of the NF's decoders ran for the agreed budget with no open crashes.
- Metrics, logs, config schema and sample config documented.
- ADRs updated; `docs/update.md` entry written; traceability table updated.
- Licence scan and SBOM clean.

7. Immediate next steps

- Confirm open questions: 11th NF, language (D1), release baseline (D3), Linux dev environment.
- Stand up the Linux dev/test environment (WSL2 or VM) with UERANSIM and Open5GS in compose; capture golden pcaps for S1–S3.
- Write ADR-0001 (language), ADR-0002 (clean-room and licensing policy), ADR-0003 (spec baseline).
- Scaffold the CMake C workspace in `libs/` and `nf/`, CI pipeline, licence scanning.
- Start `libs/crypto` and `libs/sbi` – they unblock Phase 1 (UDM) and are fully testable without any RAN.

Appendix A – Spec reference list (Rel-16 baseline)

Area	Specs
Architecture / procedures	TS 23.501, 23.502, 23.503
NAS / RAN interface	TS 24.501 (NAS), 38.413 (NGAP), 38.300/38.331 (context)
Security	TS 33.501, 35.205–208 (Milenage), 33.220
SBI (service operations)	TS 29.500, 29.501 (SBI principles), 29.502 (SMF), 29.503 (UDM), 29.507/512 (PCF), 29.510 (NRF), 29.518 (AMF), 29.509 (AUSF), 29.504/505/519 (UDR)
User plane / N4	TS 29.244 (PFCP), 29.281 (GTP-U), 29.060 (GTP background)

Appendix B – Reference tools quick guide

Tool	Role in this plan
UERANSIM	Primary UE + gNB simulator for E2E (AGPL-3.0 / commercial; test-only)
srsRAN Project + srsUE (ZMQ)	Second RAN peer for cross-validation (AGPL-3.0; test-only)
ORAN emulator (ours)	Third RAN peer; also validated against Open5GS
Open5GS	Oracle core for behaviour comparison and partial interop (AGPL-3.0; unmodified binaries, test-only)
Wireshark / tshark	Protocol analysis and golden-trace diffing
iperf3 / TRex	UPF throughput and latency